Communication Architectures

Project Report

André Alexandre: 114143 - 50%
Bernardo Marujo: 107322 - 50%

alexandreandre@ua.pt
bernardomarujo@ua.pt

Link to GitHub Repository



University of Aveiro
Department of Electronics, Telecommunications and Informatics
2025

Contents

1 Introduction **2**

2 Network Architecture and Design **2**
2.1 Addressing Plan: Dual-VTEP Strategy 3
2.2 Datacenter Switch Implementation (Linux/FRR) 3
2.3 BGP Architecture: Inter-Site Peering 4

3 Underlay Implementation: Core Connectivity **5**
3.1 Core Path Diversity (ECMP) 5

4 Client L1: MPLS Traffic Engineering **5**
4.1 Traffic Steering Strategy 5
4.2 Resilience 6
4.3 TE Database and Reservation Verification 6
4.4 RSVP Signaling Verification 7

5 Client L2: QoS (DiffServ) **7**
5.1 Classification and Marking 7
5.2 Policy Application 8
5.3 QoS Policy Verification 8

6 Results and Final Analysis **9**
6.1 Protocol Analysis: VXLAN Encapsulation 9
6.2 Simultaneous Traffic Analysis: Path Diversity and QoS 10
6.3 Control Plane Analysis: RSVP Signaling 11
6.4 Data Plane Verification: MPLS Encapsulation 12

7 Conclusion **13**

1 Introduction

The objective of this project is to perform the technical design, configuration, and testing of a double datacenter network for the operator "DC4ALL LLC". The infrastructure interconnects two geographic locations (Lisbon and Porto) via a proprietary long-range network.

The project addresses the requirements of two distinct clients:

- **Client L1:** Requires a private Ethernet network with guaranteed bandwidth (10Mbps) and high resilience.
- **Client L2:** Requires a private Ethernet network with Assured Forwarding traffic differentiation.

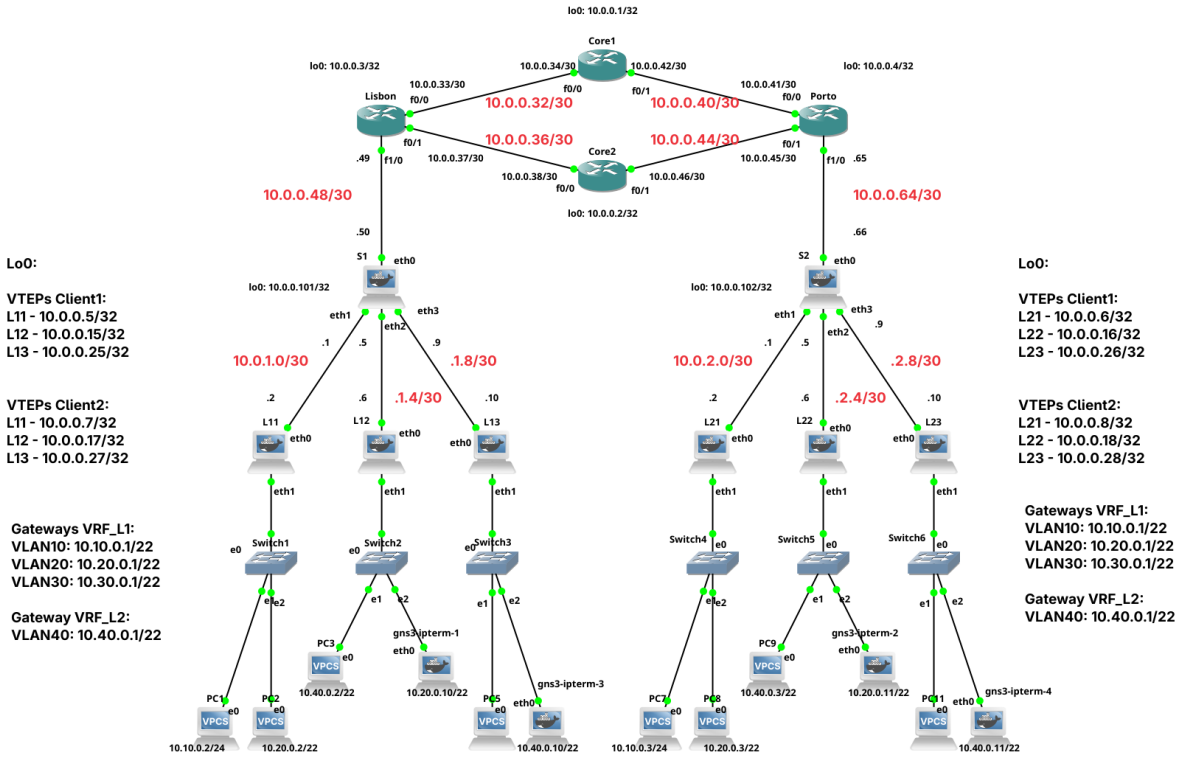


Figure 1: Network Diagram (Leaf-Spine Architecture)

2 Network Architecture and Design

The network topology consists of two Core routers (Cisco 7200) and two Datacenters. Each Datacenter implements a **Leaf-Spine** architecture using Linux FRRouting containers:

- **Spines (S1, S2):** Act as Route Reflectors and gateways to the Core network. They peer with each other to exchange routes between sites.
- **Leafs (L11-L13, L21-L23):** Act as VTEPs (VXLAN Tunnel Endpoints) and connect directly to the server racks via 802.1Q Trunks.

2.1 Addressing Plan: Dual-VTEP Strategy

To comply with the project requirements, specifically the limitation that Cisco Core routers cannot route based on VXLAN VNIs, we implemented a **Dual-VTEP** strategy. Each Leaf switch holds two distinct Loopback addresses to differentiate traffic sources at the IP layer.

- **Primary VTEP (L1):** Used for VLANs 10, 20, 30. Traffic originating from these IPs is statically routed into the **MPLS-TE Tunnel**.
- **Secondary VTEP (L2):** Used exclusively for VLAN 40. Traffic originating from these IPs follows standard OSPF routing and is subject to **DiffServ (AF21)** marking.

Device	L1 Loopback (MPLS)	L2 Loopback (DiffServ)	Role
Lisbon Router	10.0.0.3/32	-	Edge
Porto Router	10.0.0.4/32	-	Edge
Spine S1	10.0.0.101/32	-	Route Reflector
Spine S2	10.0.0.102/32	-	Route Reflector
Leaf 11	**10.0.0.5/32**	**10.0.0.7/32**	**VTEP**
Leaf 12	**10.0.0.15/32**	**10.0.0.17/32**	**VTEP**
Leaf 13	**10.0.0.25/32**	**10.0.0.27/32**	**VTEP**
Leaf 21	**10.0.0.6/32**	**10.0.0.8/32**	**VTEP**
Leaf 22	**10.0.0.16/32**	**10.0.0.18/32**	**VTEP**
Leaf 23	**10.0.0.26/32**	**10.0.0.28/32**	**VTEP**

Table 1: Dual-VTEP Addressing Plan

2.2 Datacenter Switch Implementation (Linux/FRR)

The Leafs are configured as **pure trunks**, expecting tagged traffic for all VLANs (10, 20, 30, and 40). This allows consistent handling of frames. We use the Secondary VTEP IP exclusively for the VXLAN interface associated with VLAN 40.

Terminal 2.1: Universal Leaf Configuration (Example for L11)

```
MY_VTEP_IP="10.0.0.5"
MY_VTEP_IP_L2="10.0.0.7"

ip link set eth0 up
ip addr add 10.0.1.2/30 dev eth0
ip route add default via 10.0.1.1

ip link add vrf_L1 type vrf table 1000
ip link add vrf_L2 type vrf table 2000
ip link set vrf_L1 up
ip link set vrf_L2 up

ip link add br10 type bridge
ip link set br10 master vrf_L1
ip addr add 10.10.0.1/22 dev br10
ip link set br10 up
```

```
(...)

ip link add br40 type bridge
ip link set br40 master vrf_L2
ip addr add 10.40.0.1/22 dev br40
ip link set br40 up

ip link add vni10 type vxlan local $MY_VTEP_IP dstport 4789 id 10 nolearning
ip link set vni10 master br10
ip link set vni10 type bridge_slave neigh_suppress on learning off
ip link set vni10 up

(...)

ip link add vni40 type vxlan local $MY_VTEP_IP_L2 dstport 4789 id 40 nolearning
ip link set vni40 master br40
ip link set vni40 type bridge_slave neigh_suppress on learning off
ip link set vni40 up

CABLES="eth1"

for IFACE in $CABLES; do
    ip link set $IFACE up

    ip link add link $IFACE name $IFACE.10 type vlan id 10
    ip link set $IFACE.10 master br10
    ip link set $IFACE.10 up

    (...)

    ip link add link $IFACE name $IFACE.40 type vlan id 40
    ip link set $IFACE.40 master br40
    ip link set $IFACE.40 up
done
```

2.3 BGP Architecture: Inter-Site Peering

To enable scalable control-plane learning, the Spine switches (S1 in Lisbon and S2 in Porto) are configured as **BGP Route Reflectors**. This design eliminates the need for a full-mesh BGP topology among the Leafs, which would be inefficient and difficult to manage.

To support the multi-site requirement, the two Spines establish a direct **BGP EVPN peering session** with each other. This allows MAC and IP reachability information learned in Lisbon to be propagated to Porto, enabling End-to-End connectivity.

Terminal 2.2: Spine S1 BGP Configuration (with Peer Groups)

```
router bgp 22900
 ! Define Peer Group for Local Leafs to simplify config
 neighbor LEAVES peer-group
 neighbor LEAVES remote-as 22900
 neighbor LEAVES update-source 10.0.0.101

 ! Apply Group to Local Leafs
 neighbor 10.0.0.5 peer-group LEAVES
 neighbor 10.0.0.15 peer-group LEAVES
 neighbor 10.0.0.25 peer-group LEAVES

 ! Peering with Remote Spine S2 (Inter-Site)
 neighbor 10.0.0.102 remote-as 22900
 neighbor 10.0.0.102 update-source 10.0.0.101
 !
 address-family l2vpn evpn
  ! Activate Leafs (Route Reflector Clients)
```

```
  neighbor LEAVES activate
  neighbor LEAVES route-reflector-client

  ! Activate Remote Spine (Peer - NOT a client)
  neighbor 10.0.0.102 activate
 exit-address-family
```

3 Underlay Implementation: Core Connectivity

OSPF Area 0 is configured across all devices. We ensured that the Linux Leafs advertise both Loopback addresses into OSPF so they are reachable from the Core.

3.1 Core Path Diversity (ECMP)

Traffic between datacenters is load-balanced across Core 1 and Core 2. The traceroute below confirms that the Lisbon Router utilizes multiple paths to reach Porto.

Terminal 3.1: Underlay Traceroute (OSPF ECMP)

```
Lisbon#traceroute 10.0.0.4
Type escape sequence to abort.
Tracing the route to 10.0.0.4
  1 10.0.0.34 16 msec
    10.0.0.38 12 msec
    10.0.0.34 16 msec
  2 10.0.0.45 24 msec
    10.0.0.41 36 msec
    10.0.0.45 28 msec
```

4 Client L1: MPLS Traffic Engineering

Client L1 requires guaranteed bandwidth. We configured an MPLS-TE tunnel on Lisbon and Porto routers.

4.1 Traffic Steering Strategy

We implemented Static Routing based on the **Destination VTEP IP**. Since Client L1 traffic is destined for the Primary IPs of the remote site (e.g., .6, .16, .26), we force these specific IPs into the tunnel.

Terminal 4.1: Static Routes for Traffic Steering (Lisbon)

```
! Route traffic for Porto's Client L1 VTEPs into the Tunnel
ip route 10.0.0.6 255.255.255.255 Tunnel1
ip route 10.0.0.16 255.255.255.255 Tunnel1
ip route 10.0.0.26 255.255.255.255 Tunnel1
```

Traffic destined for Client L2 (e.g., 10.0.0.8) is **not** matched by these routes and therefore follows the standard physical path.

4.2 Resilience

The tunnel is configured with `path-option 1 dynamic`. If the primary link fails, RSVP-TE automatically recalculates and signals a new path via the other Core router.

Terminal 4.2: Tunnel Status (Resilience)

```
Name: Lisbon_t1                           (Tunnel1) Destination: 10.0.0.4
  Status:
    Admin: up          Oper: up     Path: valid       Signalling: connected
    path option 1, type dynamic (Basis for Setup, path weight 2)

  Config Parameters:
    Bandwidth: 10000    kbps (Global)  Priority: 7  7   Affinity: 0x0/0xFFFF
    Metric Type: TE (default)

  Active Path Option Parameters:
    State: dynamic path option 1 is active

  InLabel  :  -
  OutLabel : FastEthernet0/1, 16
  Next Hop : 10.0.0.38
  RSVP Signalling Info:
       Src 10.0.0.3, Dst 10.0.0.4, Tun_Id 1, Tun_Instance 9
    RSVP Path Info:
      My Address: 10.0.0.37
      Explicit Route: 10.0.0.38 10.0.0.46 10.0.0.45 10.0.0.4
      Record   Route:   NONE
      Tspec: ave rate=10000 kbits, burst=1000 bytes, peak rate=10000 kbits

  History:
    Tunnel:
      Time since created: 8 hours, 4 minutes
      Time since path change: 8 hours, 3 minutes
      Number of LSP IDs (Tun_Instances) used: 9
    Current LSP: [ID: 9]
      Uptime: 8 hours, 3 minutes
```

4.3 TE Database and Reservation Verification

To ensure that the network is aware of the available bandwidth resources, OSPF was configured with MPLS-TE extensions. The output below (Listing 4.3) displays the contents of the Traffic Engineering database on the Lisbon Router.

We can observe two links with a **Maximum Reservable Bandwidth** of 1250 kbytes/sec (10 Mbps). On the link connected to `10.0.0.37` (Fragment 2), the available bandwidth at **Priority 7 is 0**. This confirms that the MPLS-TE tunnel is active on this path and has reserved the entire 10Mbps pool, strictly enforcing the SLA for Client L1.

Terminal 4.3: OSPF MPLS Traffic Engineering Database

```
  OSPF Router with ID (10.0.0.3) (Process ID 1)

  Area 0 has 2 MPLS TE links. Area instance is 4.

  Links in hash bucket 0.
    Link is associated with fragment 1. Link instance is 4
      Link connected to Broadcast network
      Link ID : 10.0.0.33
      Interface Address : 10.0.0.33
      Admin Metric te: 1 igp: 1
      Maximum bandwidth : 12500000
```

```
      Maximum reservable bandwidth : 1250000
      Number of Priority : 8
      Priority 0 : 1250000      Priority 1 : 1250000
      Priority 2 : 1250000      Priority 3 : 1250000
      Priority 4 : 1250000      Priority 5 : 1250000
      Priority 6 : 1250000      Priority 7 : 1250000
      Affinity Bit : 0x0

  Links in hash bucket 33.
    Link is associated with fragment 2. Link instance is 4
      Link connected to Broadcast network
      Link ID : 10.0.0.37
      Interface Address : 10.0.0.37
      Admin Metric te: 1 igp: 1
      Maximum bandwidth : 12500000
      Maximum reservable bandwidth : 1250000
      Number of Priority : 8
      Priority 0 : 1250000      Priority 1 : 1250000
      Priority 2 : 1250000      Priority 3 : 1250000
      Priority 4 : 1250000      Priority 5 : 1250000
      Priority 6 : 1250000      Priority 7 : 0
      Affinity Bit : 0x0
```

4.4 RSVP Signaling Verification

While OSPF distributes information about bandwidth availability, the Resource Reservation Protocol (RSVP) is responsible for the actual signaling and maintenance of the reserved path.

The output bellow (Listing 4.4) taken from the router **Core2** confirms active reservations for the MPLS-TE tunnels. We observe two distinct flows:

- **10.0.0.3 → 10.0.0.4:** The tunnel from Lisbon to Porto.
- **10.0.0.4 → 10.0.0.3:** The return tunnel from Porto to Lisbon.

In both cases, the **BPS** column shows `10M`, confirming that the requested 10 Mbps have been allocated on the physical interfaces, guaranteeing the SLA for Client L1.

Terminal 4.4: RSVP Reservation State (Core 2)

```
Core2#show ip rsvp reservation
To            From          Pro DPort Sport Next Hop      I/F     Fi Serv BPS
10.0.0.3      10.0.0.4      0   1     12    10.0.0.37     Fa0/0    SE LOAD 10M
10.0.0.4      10.0.0.3      0   1     9     10.0.0.45     Fa0/1    SE LOAD 10M
```

5 Client L2: QoS (DiffServ)

Client L2 traffic traverses the physical links but requires prioritized handling (AF21).

5.1 Classification and Marking

Since standard routing is used, we apply an ACL on the **Ingress Interface** of the Edge Router. We match traffic originating from our local **Secondary VTEP IPs**.

Terminal 5.1: QoS Classification ACL (Lisbon)

```
! Match traffic originating from Local Client L2 Leaves
access-list 105 permit udp host 10.0.0.7 any eq 4789
access-list 105 permit udp host 10.0.0.17 any eq 4789
access-list 105 permit udp host 10.0.0.27 any eq 4789
```

5.2 Policy Application

The marking policy is applied ingress from the Datacenter. The queuing policy is applied egress towards the Core.

Terminal 5.2: DiffServ Policy Application

```
! Ingress: Mark packets as AF21
policy-map MARK_L2_INGRESS
 class CLASS_L2_VXLAN
  set ip dscp af21

! Egress: Guarantee Bandwidth for AF21
policy-map QOS_CORE_EGRESS
 class CLASS_AF21
  bandwidth 10000
```

5.3 QoS Policy Verification

To validate the Differentiated Services implementation, we examined the policy-map statistics on the Lisbon Edge Router.

Listing 5.3 shows the **Ingress Interface** (f1/0), which connects to the Datacenter Spine. We can observe that 330 packets matched the Access Control List (ACL 105) defining Client L2 traffic. The **"Packets marked"** counter confirms that the router applied the **DSCP AF21** tag to this specific flow.

Terminal 5.3: Ingress Marking Verification (f1/0)

```
Lisbon#show policy-map interface f1/0
 FastEthernet1/0

  Service-policy input: MARK_L2_INGRESS

    Class-map: CLASS_L2_VXLAN (match-all)
      330 packets, 40723 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: access-group 105
      QoS Set
        dscp af21
          Packets marked 330
```

Listing 5.4 shows the **Egress Interface** (f0/0), which connects to the Provider Core. Here, the policy matches traffic based on the DSCP AF21 tag applied previously. The output confirms that the `CLASS_AF21` queue is active and has enforced the configured **bandwidth guarantee of 10000 kbps**.

Terminal 5.4: Egress Queuing Verification (f0/0)

```
Lisbon#show policy-map interface f0/0
 FastEthernet0/0

  Service-policy output: QOS_CORE_EGRESS

    Class-map: CLASS_AF21 (match-all)
      120 packets, 14400 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: ip dscp af21 (18)
      Queueing
      queue limit 64 packets
      (queue depth/total drops/no-buffer drops) 0/0/0
      (pkts output/bytes output) 120/14400
      bandwidth 10000 kbps
```

6 Results and Final Analysis

6.1 Protocol Analysis: VXLAN Encapsulation

To validate the correct operation of the Overlay network, we performed a packet capture using Wireshark. Figure 2 captures an **iPerf3** throughput test between Client L1 nodes located in different datacenters.

The capture clearly demonstrates the nested encapsulation structure required for the solution:

1. **Underlay (Outer Header):** The Internet Protocol Version 4 header shows the source as `10.0.0.16` (Leaf 22 in Porto) and the destination as `10.0.0.15` (Leaf 12 in Lisbon). This confirms that the VTEPs are routing traffic across the WAN.

2. **VXLAN Header:** The UDP destination port **4789** identifies the traffic as VXLAN. The header is inserted between the outer UDP and inner Ethernet frames.

3. **Overlay (Inner Header):** The inner IPv4 header reveals the actual client traffic: Source `10.20.0.11` to Destination `10.20.0.10`. This corresponds to the private address space of **Client L1 (VLAN 20)**, which is completely invisible to the Core routers.

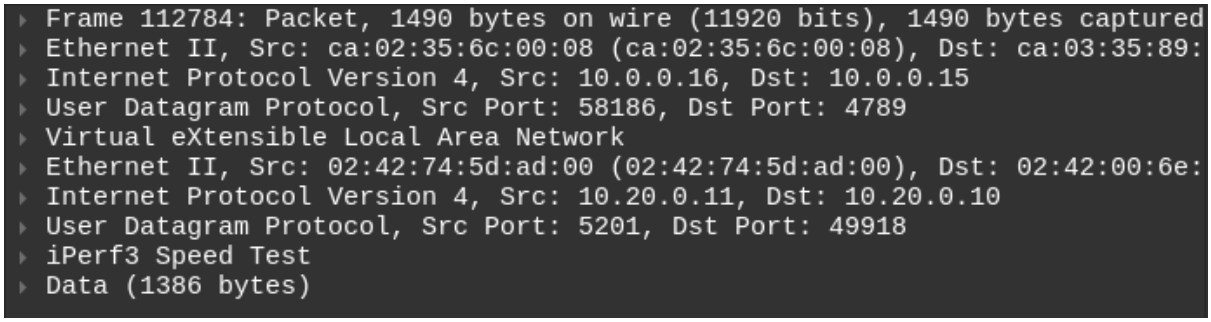


Figure 2: Wireshark capture showing VXLAN encapsulation of Client L1 iPerf3 traffic.

6.2 Simultaneous Traffic Analysis: Path Diversity and QoS

To definitively validate the isolation and traffic steering mechanisms, we performed simultaneous **iPerf3** download tests for both clients. Figure 3 presents a split-screen view of the results, with Client 1 on the left and Client 2 on the right.

This capture provides conclusive evidence of the "Dual-VTEP" architecture:

1. **Traffic Steering (Different Cores):**
    - **Left Side (Client 1):** The packet capture window title shows traffic captured on **Core 2**. The Outer Source IP is `10.0.0.16` (Primary VTEP). This confirms that Client 1 traffic was steered into the MPLS-TE tunnel path.
    - **Right Side (Client 2):** The packet capture window title shows traffic captured on **Core 1**. The Outer Source IP is `10.0.0.28` (Secondary VTEP). This confirms that Client 2 traffic bypassed the tunnel and followed the standard OSPF routing path.
2. **DSCP Marking Verification:**
    - **Client 1 (Left):** The Differentiated Services Field is **0x00 (CS0/Default)**. This is expected, as Client 1 relies on MPLS-TE bandwidth reservation rather than DiffServ IP prioritization.
    - **Client 2 (Right):** The Differentiated Services Field is clearly marked as **0x48 (AF21)**. This proves that our Ingress ACL on the Edge Router identified the Secondary VTEP source IP and applied the correct QoS tag for the Core network.

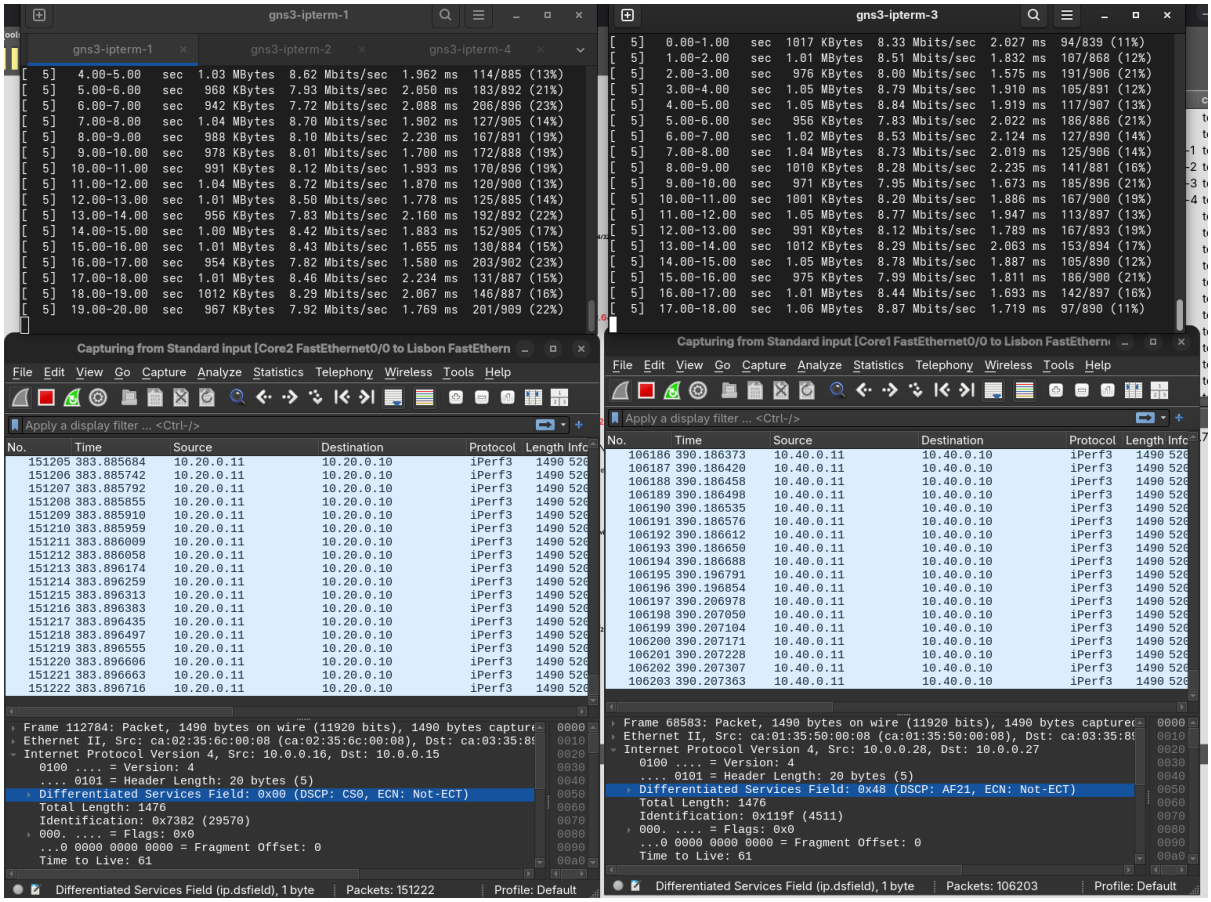


Figure 3: Simultaneous iPerf3 tests showing Path Diversity (Core 1 vs Core 2) and QoS Marking (AF21 vs CS0).

6.3 Control Plane Analysis: RSVP Signaling

We analyzed the signaling traffic to verify how the MPLS-TE tunnel is established and maintained. Figure 4 shows a packet capture taken specifically on the link between the routers Lisbon and Core2.

The presence of **RSVP** messages on this specific link confirms that the tunnel is currently traversing Core 2. If the tunnel were using Core 1, these messages would not be present here.

Key observations from the **RSVP PATH Message**:

1. **Dynamic Path Calculation (Explicit Route):** The `EXPLICIT ROUTE` object lists the sequence of interface IP addresses that the tunnel must traverse: `10.0.0.38` $\rightarrow$ `10.0.0.46` $\rightarrow$ `10.0.0.45`. These IPs correspond to the physical path through **Core 2**, showing that OSPF dynamically calculated this path and RSVP is actively pinning it.

2. **Bandwidth Reservation (TSpec):** The `SENDER TSPEC` field contains the Traffic Specification parameters requested by the Head-end router (Lisbon). The Token Bucket rate is set to **1,250,000 bytes/sec**.

$$1,250,000 \text{ bytes/s} \times 8 \text{ bits/byte} = 10,000,000 \text{ bps} = 10 \text{ Mbps}$$

This confirms that the 10 Mbps bandwidth requirement defined in the project configuration is being signaled on the wire and reserved at every hop along the path.

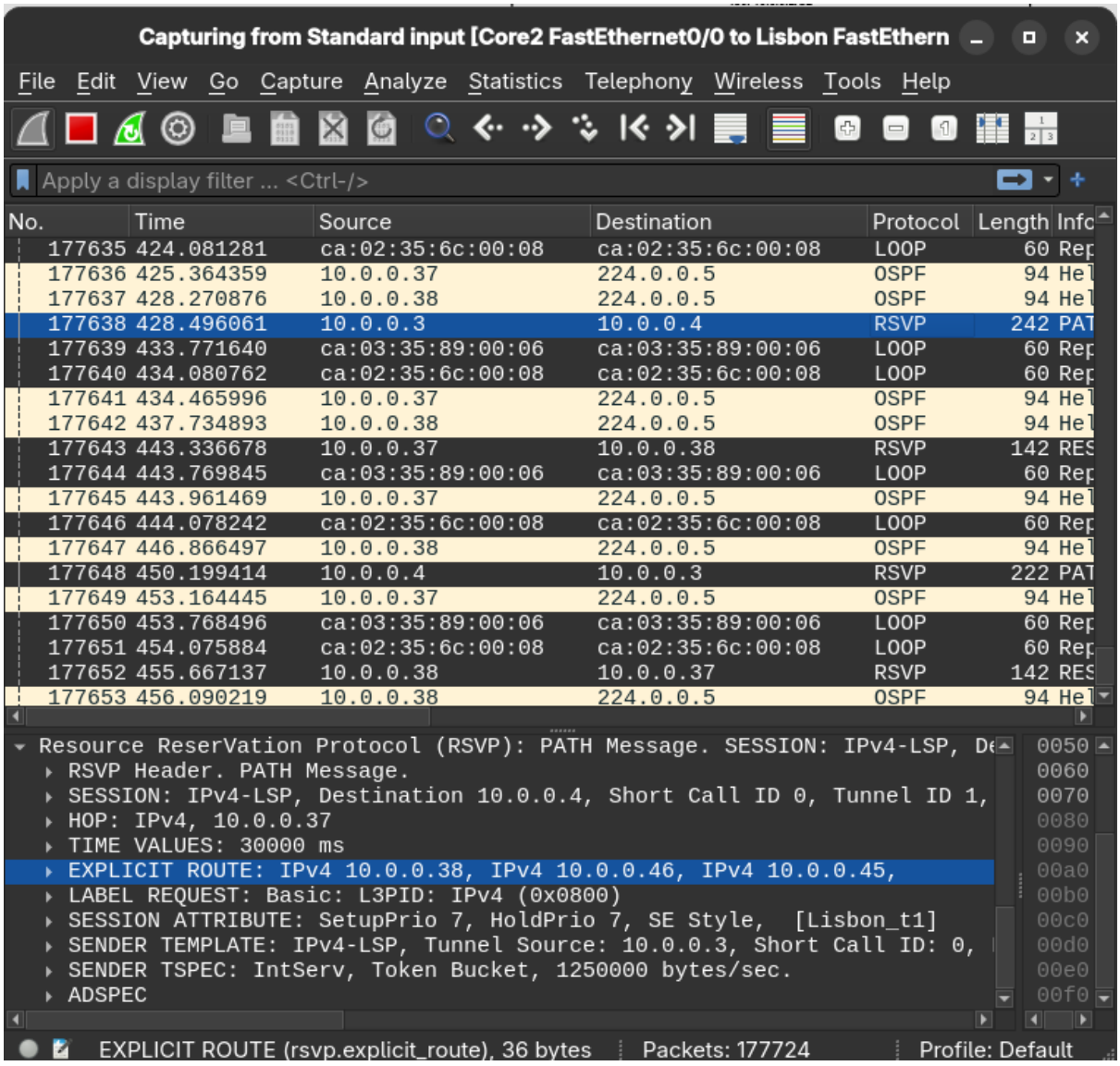


Figure 4: RSVP PATH message captured on the Core 2 link, showing the Explicit Route via Core 2 and the 10Mbps TSpec reservation.

6.4 Data Plane Verification: MPLS Encapsulation

While the RSVP capture proved that the Control Plane signaled the reservation, we needed to verify that user traffic was actually being encapsulated with MPLS labels in the Data Plane.

Figure 5 shows a captured ICMP (Ping) traversing the Core network. This packet represents the complete protocol stack utilized in our solution:

1. **MPLS Header (Label 17):** The presence of the MPLS header is the critical differentiator. The label **17** is pushed onto the packet by the router. This label tells

the intermediate routers to switch the packet along the pre-reserved path (Tunnel 1) rather than performing a standard IP lookup.

2. **Outer IP (VTEP to VTEP):** Below the MPLS label, we see the IPv4 header with Source `10.0.0.5` and Destination `10.0.0.6`. These are the Primary Loopback addresses of the Leaf switches, confirming that this is Client L1 traffic which we explicitly steered into the tunnel.

3. **VXLAN Overlay:** The UDP header (Dst Port 4789) and VXLAN header confirm that the payload is an Ethernet frame from the datacenter overlay.

4. **Inner Payload (Client Traffic):** At the bottom of the stack, we see the actual user traffic: An ICMP request from `10.10.0.2` to `10.10.0.3`.

This capture conclusively proves that Client L1 traffic is being tunneled through the MPLS-TE path, fully isolated from the underlying physical topology.

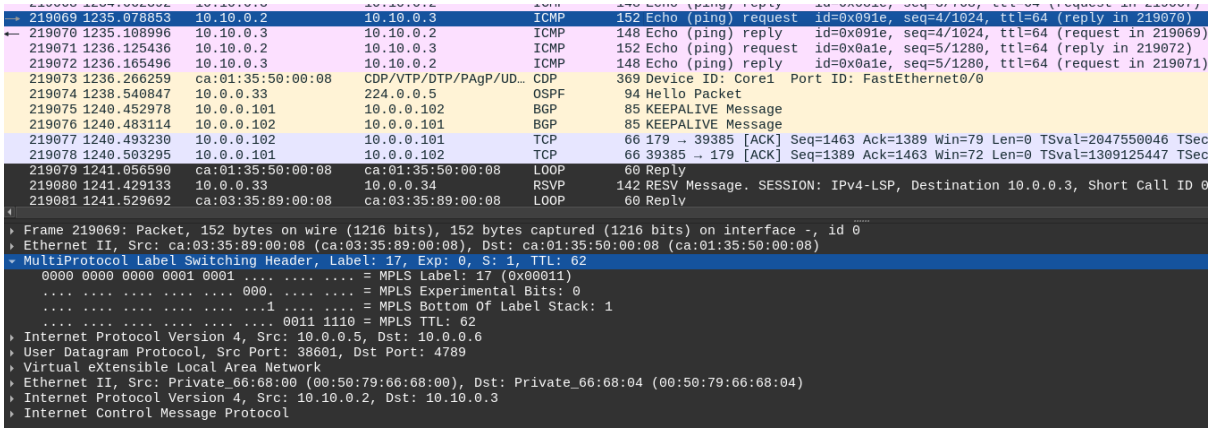


Figure 5: Data Plane capture showing the 4-layer encapsulation: MPLS Label 17 → Outer IP → VXLAN → Inner ICMP.

7 Conclusion

The implemented Leaf-Spine architecture successfully meets all requirements:

- **Isolation:** Complete separation of Client L1 and L2 traffic.
- **Scalability:** BGP EVPN eliminates flooding and allows efficient MAC learning across datacenters.
- **Client L1:** Secured with MPLS-TE (10Mbps reserved path) and VRF isolation via VXLAN VNIs 10, 20, 30.
- **Client L2:** Secured with VNI 40 and differentiated via DiffServ (AF21) QoS policies in the core.